

WEEK 5

Healthcare Data Cleaning & Data Modeling

Power Query & Power BI Data Modeling Project Report

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1. Problem Statement

A healthcare organization wants to analyze patient information, medical conditions, admissions, and healthcare-related characteristics. The raw data, exported from the organization's records system, arrives as a single flat table containing patient demographics, admission details, and billing information all mixed together. Before any meaningful clinical analytics or reporting can take place, this raw data must be cleaned, validated, and restructured into a proper data model suitable for an interactive Power BI report.

This project walks through that end-to-end process: inspecting the raw data, cleaning and standardizing it in Power Query, splitting it into logical dimension and fact tables, and finally establishing relationships between those tables so the dataset is ready for clinical analytics and dashboarding.

2. Objectives

- Import and inspect patient records, admission types, and billing fields in Power Query.
- Handle null values in diagnostic codes and standardize categorical text attributes.
- Create derived conditional columns (e.g., Age Group buckets: 0-18, 19-35, 36-60, 60+).
- Structure dimension tables (Dim_Patient, Dim_Admission) and fact tables (Fact_Billing).

3. Tasks & Deliverables

- Group patient ages into logical demographic categories via Power Query.
- Standardize admission types (Emergency, Elective, Urgent).
- Set up star-schema model relationships between patient dimensions and admission facts.
- Output a cleaned data model ready for clinical analytics.

4. Dataset Overview

The source dataset, Healthcare_Raw, contains over 54,900 rows and combines three conceptual entities in one table: patient information (Name, Age, Gender, Blood Type), admission information (Hospital, Doctor, Medical Condition, Room

Number, Admission Type, Medication, Test Results, Date of Admission, Discharge Date), and billing information (Insurance Provider, Billing Amount). Keeping all of this in a single flat table makes the data difficult to filter, aggregate, and relate correctly, and it introduces redundancy since patient and hospital details repeat across every admission record for the same person.

The goal of the transformation work described in this report is to break this flat table apart into a proper star schema: one fact table (Billing) that records each billing transaction, and two dimension tables (Dim_Patient and Dim_Admission) that describe the entities involved in each transaction without repeating their attributes row after row.

5. Data Cleaning & Transformation Steps

5.1 Initial Cleaning of Healthcare_Raw

Before any structural changes were made, the raw import was cleaned in Power Query using a standard sequence of transformations: promoting the first row to headers, correcting column data types, trimming stray whitespace from text fields, cleaning non-printable characters, removing duplicate rows, and capitalizing each word in categorical text columns (such as Medical Condition, Admission Type, and Medication) so that inconsistent capitalization did not create false distinct categories. A conditional column was also added at this stage to support later categorization needs, such as Age Group bucketing.

5.1.1 Building the Dim_Admission Table

With the base cleaning applied, attention turned to isolating the admission-related attributes into their own dimension table. Columns that belonged to the patient or billing tables (Name, Age, Gender, Blood Type, Date of Admission, Insurance Provider, Billing Amount) were removed from this query, leaving only the columns that describe an admission event. An Index Column was then added, starting at 1, to generate a unique Admission id — this surrogate key is what allows the admission dimension to be referenced from the fact table later, without relying on any single business column being unique on its own.

File Home Transform Add Column View Tools Help

Column From Custom Invoke Custom Examples Column Function General

Conditional Column Index Column Merge Columns Format Parse

From 0 From 1 From Text From Number From Date & Time

Queries [3] Healthcare_Raw Dim_Patient Dim_Admission

Create a new column with an index starting at 1. (#"Cleaned Text1", {"lane", "Age", "Gender", "Blood Type", "Date of Admission", "Insurance Provider", "Billing Amount",

	Medical Condition	Doctor	Room Number	Admission Type	Medication	Test Results
1	Cancer	Matthew Smith	328	Urgent	Paracetamol	Normal
2	Obesity	Samantha Davies	265	Emergency	Ibuprofen	Inconclusive
3	Obesity	Tiffany Mitchell	205	Emergency	Aspirin	Normal
4	Vang, Diabetes	Kevin Wells	450	Elective	Ibuprofen	Abnormal
5	Cancer	Kathleen Hanna	458	Urgent	Penicillin	Abnormal
6	Asthma	Taylor Newton	389	Urgent	Ibuprofen	Normal
7	Diabetes	Kelly Olson	389	Emergency	Paracetamol	Inconclusive
8	aldez, Cancer	Suzanne Thomas	277	Emergency	Paracetamol	Inconclusive
9	Asthma	Daniel Ferguson	316	Elective	Aspirin	Abnormal
10	Cancer	Heather Day	249	Elective	Paracetamol	Inconclusive
11	Cancer	John Duncan	394	Urgent	Paracetamol	Normal
12	Hypertension	Douglas Mayo	288	Urgent	Lipitor	Inconclusive
13	res Diabetes	Kenneth Fletcher	134	Emergency	Penicillin	Abnormal
14	Asthma	Theresa Freeman	309	Urgent	Lipitor	Normal
15	Cancer	Roberta Stewart	182	Urgent	Paracetamol	Normal
16	Obesity	Maria Dougherty	465	Elective	Aspirin	Inconclusive
17	Arthritis	Erica Spencer	114	Elective	Paracetamol	Normal
18	ones Obesity	Justin Kim	449	Urgent	Paracetamol	Abnormal
19	Hypertension	Justin Moore Jr.	260	Elective	Aspirin	Abnormal
20	Asthma	Denise Galloway	465	Elective	Penicillin	Normal
21	Asthma	Krista Smith	115	Elective	Aspirin	Normal
22	Asthma	Gregory Smith	295	Urgent	Lipitor	Normal
23	Arthritis	Vanessa Newton	327	Urgent	Lipitor	Inconclusive
24	Asthma	Donna Martinez Md	119	Urgent	Lipitor	Inconclusive
25	Arthritis	Arthritis	100	Emergency	Paracetamol	Inconclusive

7 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 10:50

Query Settings

PROPERTIES

Name Dim_Admission

APPLIED STEPS

Source

Promoted Headers

Changed Type

Trimmed Text

Cleaned Text

Reordered Columns

Removed Duplicates

Capitalized Each Word

Changed Type1

Added Conditional Column

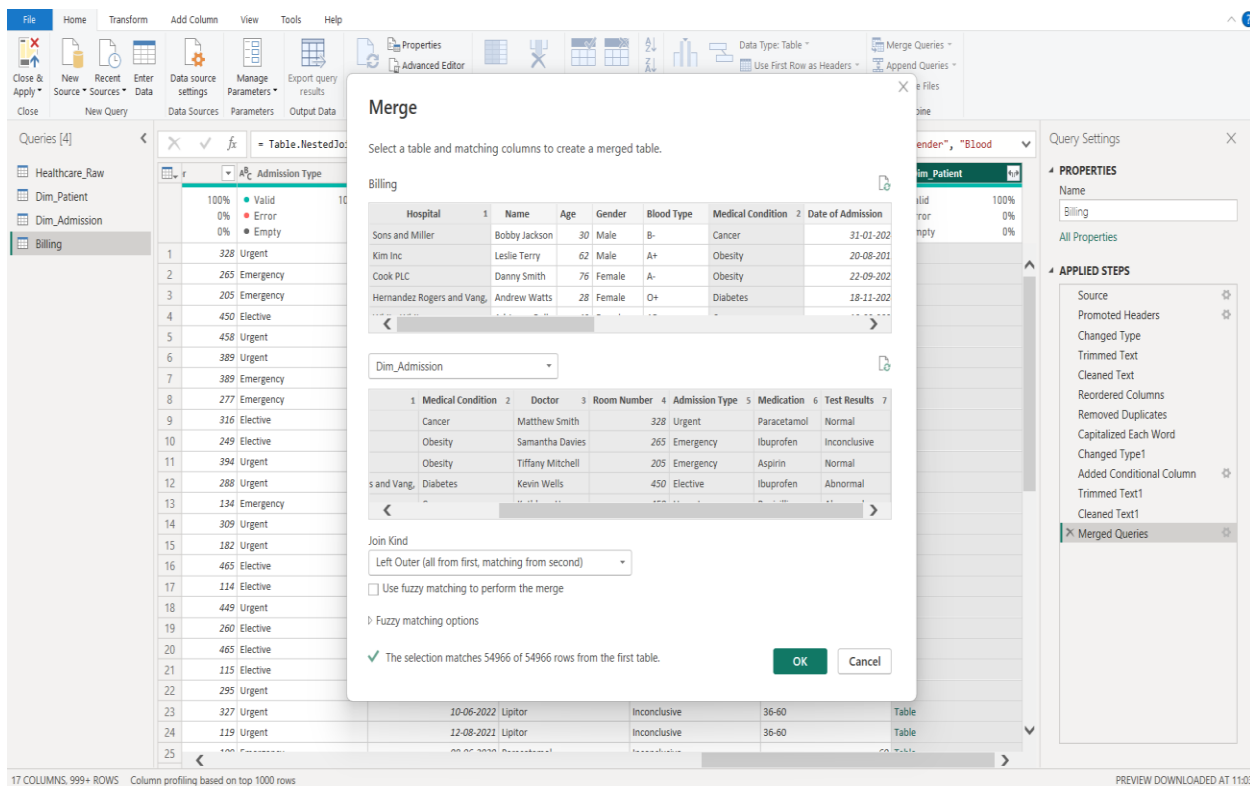
Trimmed Text1

Cleaned Text1

Removed Columns

5.1.2 Adding a Hospital Column and Reordering

A Hospital column was generated for the admission dimension, and the columns were reordered into a clean, logical sequence: Admission id, Hospital, Medical Condition, Doctor, Room Number, Admission Type, and Medication. Reordering columns at this stage is a small step, but it keeps the query readable and makes the resulting table easier to review and validate before it is used elsewhere in the model.



5.1.4 Expanding the Dim_Patient Reference

A second merge brought Dim_Patient into the Billing query. Rather than expanding all of the patient columns into Billing, only the Patient id column was selected in the expand dialog. This is a deliberate modeling choice: expanding Name, Age, Gender, Blood Type, or Age Group into the fact table would duplicate patient attributes across every billing row for that patient, undermining the point of having a separate patient dimension. Keeping only the key column preserves a clean foreign-key relationship instead.

File Home Transform Add Column View Tools Help

Close & Apply * New Source * Recent Sources * Enter Data Data source settings Data Sources Parameters Output Data Query Properties Advanced Editor Manage * Choose Columns * Remove Columns * Keep Rows * Remove Rows * Split Column * Group By Data Type: Table * Use First Row as Headers * Merge Queries * Append Queries * Combine Files * Combine

Queries [4] Healthcare_Raw Dim_Patient Dim_Admission Billing

Table.ExpandTableColumn(#Removed Columns, "Dim_Patient", {"Patient id"}, {"Dim_Patient.Patient id"})

	Date of Admission	Insurance Provider	Billing Amount	Discharge Date	Dim_Patient.Patient id	Dim_Admission
1	31-01-2024	Blue Cross	18856.28131	02-02-202		
2	20-08-2019	Medicare	33643.32729	26-08-201		
3	22-09-2022	Aetna	27955.09608	07-10-202		
4	18-11-2020	Medicare	37909.78241	18-12-200		
5	19-09-2022	Aetna	14238.31781	09-10-202		
6	20-12-2023	UnitedHealthcare	48145.11095	24-12-202		
7	03-11-2020	Medicare	19580.87234	15-11-202		
8	28-12-2021	Cigna	45820.46272	07-01-202		
9	01-07-2020	Cigna	50119.22279	14-07-200		
10	23-05-2021	UnitedHealthcare	19784.63106	22-06-202		
11	19-04-2020	Medicare	12576.79561	22-04-202		
12	13-08-2023	Medicare	7999.58688	05-09-202		
13	12-12-2019	Cigna	43282.28336	28-12-201		
14	22-05-2020	UnitedHealthcare	33207.70663	19-06-200		
15	08-10-2021	UnitedHealthcare	40701.59923	13-10-2021		
16	01-01-2023	Blue Cross	12263.35743	11-01-2023		
17	23-06-2020	UnitedHealthcare	24499.8479	14-07-2020		
18	08-03-2020	Cigna	17440.46544	02-04-2020		
19	04-03-2021	Blue Cross	18843.02302	14-03-2021		
20	15-11-2022	Cigna	23762.20358	22-11-2022		
21	28-06-2023	Blue Cross	42.51458855	02-07-2023		
22	21-01-2020	Aetna	17695.91162	09-02-2020		
23	12-05-2022	Aetna	5998.102908	10-06-2022		
24	02-08-2021	Aetna	25250.05243	12-08-2021		
25	15-05-2020	Medicare	33211.29542	08-06-2020		

6 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 11:06

Query Settings

PROPERTIES

Name

Billing

APPLIED STEPS

Source

Promoted Headers

Changed Type

Trimmed Text

Cleaned Text

Reordered Columns

Removed Duplicates

Capitalized Each Word

Changed Type1

Added Conditional Column

Trimmed Text1

Cleaned Text1

Merged Queries

Merged Queries1

Removed Columns

Expanded Dim_Patient

File Home Transform Add Column View Tools Help

Close & Apply * New Source * Recent Sources * Enter Data Data source settings Data Sources Parameters Output Data Query Properties Advanced Editor Manage * Choose Columns * Remove Columns * Keep Rows * Remove Rows * Split Column * Group By Data Type: Table * Use First Row as Headers * Merge Queries * Append Queries * Combine Files * Combine

Queries [4] Healthcare_Raw Dim_Patient Dim_Admission Billing

Table.ExpandTableColumn(#Removed Columns, "Dim_Admission", {"Admission id"}, {"Admission id"})

	Date of Admission	Insurance Provider	Billing Amount	Discharge Date	Dim_Patient	Admission id
1	31-01-2024	Blue Cross	18856.			1
2	20-08-2019	Medicare	33643.			2
3	22-09-2022	Aetna	27955.1			3
4	18-11-2020	Medicare	37909.			4
5	19-09-2022	Aetna	14238.			5
6	19-09-2022	Aetna	14238.			50130
7	19-09-2022	Aetna	14238.			5
8	19-09-2022	Aetna	14238.			50130
9	20-12-2023	UnitedHealthcare	48145.			6
10	03-11-2020	Medicare	19580.			7
11	28-12-2021	Cigna	45820.46272	07-01-2022		8
12	01-07-2020	Cigna	50119.22279	14-07-2020		9
13	23-05-2021	UnitedHealthcare	19784.63106	22-06-2021		10
14	19-04-2020	Medicare	12576.79561	22-04-2020		11
15	13-08-2023	Medicare	7999.58688	05-09-2023		12
16	12-12-2019	Cigna	43282.28336	28-12-2019		13
17	22-05-2020	UnitedHealthcare	33207.70663	19-06-2020		14
18	08-10-2021	UnitedHealthcare	40701.59923	13-10-2021		15
19	01-01-2023	Blue Cross	12263.35743	11-01-2023		16
20	23-06-2020	UnitedHealthcare	24499.8479	14-07-2020		17
21	08-03-2020	Cigna	17440.46544	02-04-2020		18
22	04-03-2021	Blue Cross	18843.02302	14-03-2021		19
23	15-11-2022	Cigna	23762.20358	22-11-2022		20
24	28-06-2023	Blue Cross	42.51458855	02-07-2023		21
25	21-01-2020	Aetna	17695.91162	09-02-2020		22

6 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 11:07

Query Settings

PROPERTIES

Name

Billing

APPLIED STEPS

Source

Promoted Headers

Changed Type

Trimmed Text

Cleaned Text

Reordered Columns

Removed Duplicates

Capitalized Each Word

Changed Type1

Added Conditional Column

Trimmed Text1

Cleaned Text1

Merged Queries

Merged Queries1

Removed Columns

Expanded Dim_Admission

5.1.5 Finalizing the Billing Fact Table

After both merges and expansions are applied, the Billing table was reduced to six columns: Date of Admission, Insurance Provider, Billing Amount, Discharge Date, Patient id, and Admission id. Of these, Patient id and Admission id are

foreign keys — they contain no descriptive information themselves, but they link every billing transaction back to exactly one patient and exactly one admission record. This is the defining shape of a fact table: numeric and transactional measures (Billing Amount, dates) alongside the foreign keys needed to join out to the dimensions that give those measures context.

The screenshot displays the Microsoft Power BI Desktop interface. The main view shows a data table with the following columns: Date of Admission, Insurance Provider, Billing Amount, Discharge Date, Patient id, and Admission id. The table contains 25 rows of data. The interface includes a ribbon with various data transformation tools and a right-hand pane with Properties and Applied Steps sections.

	Date of Admission	Insurance Provider	Billing Amount	Discharge Date	Patient id	Admission id
1	31-01-2024	Blue Cross	18856.28131	02-02-2024	1	1
2	20-08-2019	Medicare	33643.32729	26-08-2019	2	2
3	22-09-2022	Aetna	27955.09608	07-10-2022	3	3
4	18-11-2020	Medicare	37909.78241	18-12-2020	4	4
5	19-09-2022	Aetna	14238.31781	09-10-2022	5	5
6	19-09-2022	Aetna	14238.31781	09-10-2022	5	50130
7	20-12-2023	UnitedHealthcare	48145.11095	24-12-2023	6	6
8	19-09-2022	Aetna	14238.31781	09-10-2022	50130	5
9	19-09-2022	Aetna	14238.31781	09-10-2022	50130	50130
10	03-11-2020	Medicare	19580.87234	15-11-2020	7	7
11	28-12-2021	Cigna	45820.46272	07-01-2022	8	8
12	01-07-2020	Cigna	50119.22279	14-07-2020	9	9
13	23-05-2021	UnitedHealthcare	19784.63106	22-06-2021	10	10
14	19-04-2020	Medicare	12576.79561	22-04-2020	11	11
15	13-08-2023	Medicare	7999.58688	05-09-2023	12	12
16	12-12-2019	Cigna	43282.28336	28-12-2019	13	13
17	22-05-2020	UnitedHealthcare	33207.70663	19-06-2020	14	14
18	08-10-2021	UnitedHealthcare	40701.59923	13-10-2021	15	15
19	01-01-2023	Blue Cross	12263.35743	11-01-2023	16	16
20	23-06-2020	UnitedHealthcare	24499.8479	14-07-2020	17	17
21	08-03-2020	Cigna	17440.46544	02-04-2020	18	18
22	04-03-2021	Blue Cross	18843.02302	14-03-2021	19	19
23	15-11-2022	Cigna	23782.20358	22-11-2022	20	20
24	28-06-2023	Blue Cross	42.51458855	02-07-2023	21	21
25	21-01-2020	Aetna	17695.91162	09-02-2020	22	22

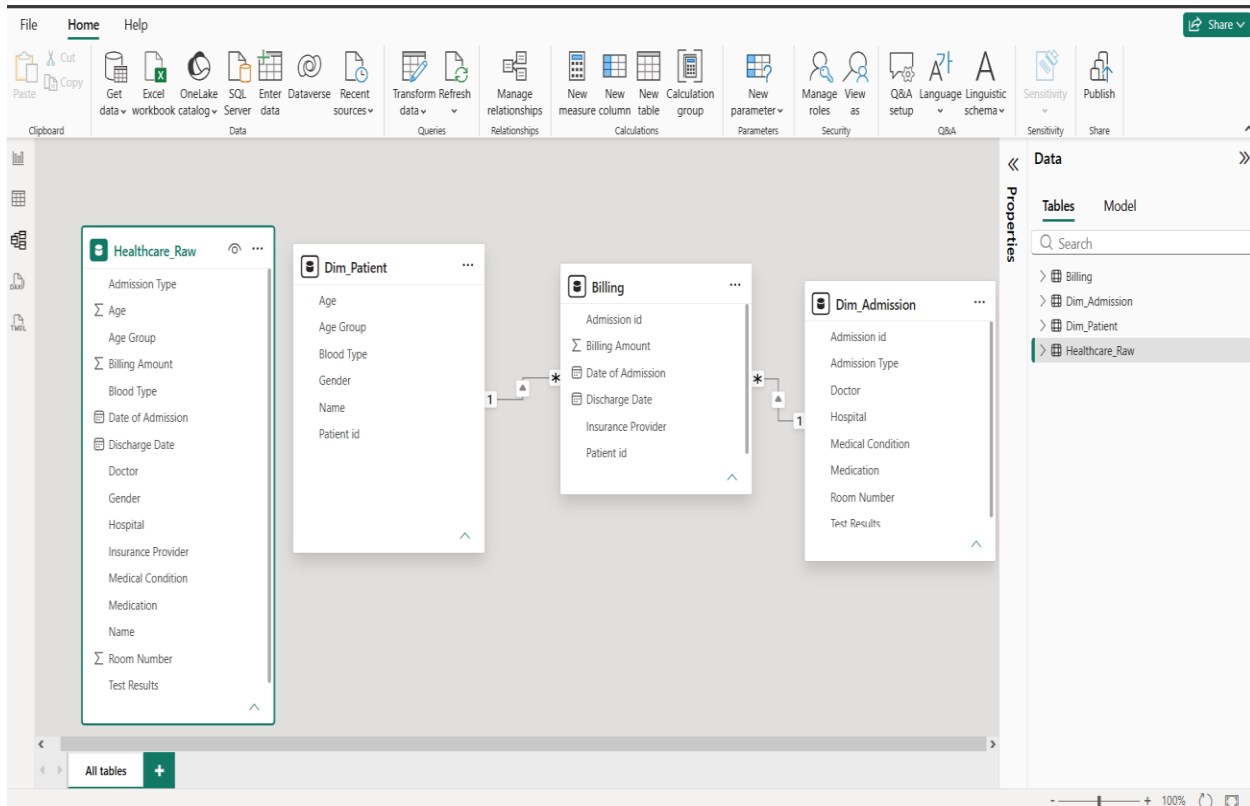
6. Data Model & Relationships

With Healthcare_Raw, Dim_Patient, Dim_Admission, and Billing all cleaned and loaded into Power BI, relationships were established in Model view to formally connect the tables. Two one-to-many relationships were created, each flowing from a dimension table's key (the "one" side) to the matching foreign key column in Billing (the "many" side):

From Table	To Table	Key / Cardinality
Dim_Patient	Billing	Patient id (1 to many)
Dim_Admission	Billing	Admission id (1 to many)

This produces a clean star schema, with Billing sitting at the center as the single fact table and Dim_Patient and Dim_Admission radiating outward as dimension

tables. A star schema like this is the standard pattern for BI reporting because it keeps calculations fast and unambiguous: measures such as total billing amount can be sliced by any patient or admission attribute through a single join hop, without the risk of double-counting that can occur in more tangled, snowflaked, or fully normalized schemas.



7. Data Quality & Validation Summary

Power Query's column quality and column distribution indicators were used throughout the cleaning process to verify the health of each field before it was carried forward into the model. Across the key admission attributes, the data consistently showed full validity with no errors or empty values, confirming that the earlier cleaning steps (trimming, capitalizing, de-duplicating) had resolved the data quality issues present in the original raw export:

Column	Valid %	Error / Empty %
Medical Condition	100%	0% / 0%
Doctor	100%	0% / 0%
Room Number	100%	0% / 0%
Admission Type	100%	0% / 0%
Medication	100%	0% / 0%
Test Results	100%	0% / 0%

Maintaining this level of visibility into column health at every stage of the query made it possible to catch inconsistencies early — for example, inconsistent capitalization of categorical values such as medical conditions or admission types — before they could propagate into the dimension and fact tables and silently break downstream aggregations in the Power BI report.

8. Challenges & Key Learnings

- Separating a single flat export into dimension and fact tables required carefully deciding which columns belonged to which entity (patient vs. admission vs. billing) before any keys could be generated.
- Because the raw data had no natural unique identifier for an admission, a surrogate key (the Index Column) had to be generated to serve as the Admission id, which then became the basis for every later merge and relationship.
- Merging on multiple descriptive columns (rather than a single key) was necessary early on since Billing and Dim_Admission did not yet share a numeric key — this made it important to verify the match rate (54,966/54,966) to be confident no rows were silently dropped.
- Care was needed when expanding merged tables to pull through only the foreign key column, rather than every column, to avoid re-introducing the redundancy the dimensional model was meant to eliminate.

9. Recommendations & Future Work

- Build out the Power BI report layer on top of this model, using the Age Group and Admission Type categorical columns as primary slicers.
- Add explicit measures (e.g., total Billing Amount, average length of stay using Date of Admission and Discharge Date) using DAX, now that the star schema is in place.

- Consider adding a Date dimension table to support time-intelligence calculations (month-over-month, year-over-year trends) in future iterations.
- Periodically re-profile column quality after any refresh of the underlying raw data to catch new data quality issues before they reach the report.

10. Conclusion

The raw healthcare dataset was successfully cleaned and restructured into a star-schema data model. Admission details were separated into Dim_Admission, patient demographics into Dim_Patient, and transactional billing information into the Billing fact table, linked via surrogate keys (Admission id and Patient id). Column-level validation throughout the process confirmed 100% data validity across the key admission attributes, with no errors or empty values remaining. This structure eliminates redundancy, supports efficient filtering and aggregation, and provides a solid, analytics-ready foundation for building the interactive Power BI report requested in the project brief.